

WELL COMPLETION REPORT (SYNTHETIC)

OIL-NHK-001 - Nahorkatiya-001

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000002
Field / District	Nahorkatiya / Dibrugarh
Block	PEL-ASSAM-02
Basin	Assam-Arakan
Formation	Girujan (GRJ)
Spud / Completion	2014-05-07 / 2014-08-20
Measured / TVD	1967 m / 1945 m
Trajectory	Vertical; inc 0.6 deg; az 197.1 deg
Formation Interval	1103-1701 m
Lithology	Argillaceous sandstone and shale
Pore Pressure Reference	2024 psi
Fracture Gradient	14.30 ppg
Mud System	Low-Solids WBM
NWIS Well Risk	Low (38.4/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	78	Conductor
13 3/8 in casing	491	Surface
9 5/8 in casing	1140	Intermediate
7 in liner/casing	1730	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2014-06-12 782 m	Fishing Operation	Critical	Downhole tool remained unrecovered after freeing attempts. Mitigation: Review BHA failure mode before resuming drilling. Lesson: Early stuck-pipe escalation may reduce probability of fishing.
2014-07-09 1259 m	Torque Spike	Critical	Sharp torque excursion occurred with RPM instability. Mitigation: Monitor torque derivative rather than absolute torque alone. Lesson: Use offset torque trend to set adaptive operating envelope.
2014-05-20 352 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2014-05-22 402 m	Stuck Pipe	High	Hookload became erratic during pipe movement. Mitigation: Apply approved freeing procedure and avoid excessive overpull. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2014-05-23 422 m	Stuck Pipe	High	Rotary movement deteriorated progressively. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.
2014-05-27 500 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Track flow balance and pit volume trend before losses become severe.
2014-06-01 595 m	Pressure Spike	High	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Check solids loading and BHA hydraulics before drilling ahead. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.

Completion Summary

The synthetic well reached 1967 m MD in the Girujan interval. The NWIS knowledge record links 15 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.